StaffSync

Modern Human Resource Management Solution

Project Report

MERN Stack: MongoDB, Express.js, React.js, Node.js

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GitHub: github.com/en24cs3010799-cmd/staffsync-hrms

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1. Introduction

StaffSync is a comprehensive web-based application built on the MERN stack (MongoDB, Express.js, React.js and Node.js) that streamlines and automates HR processes inside an organization. It provides distinct roles and access levels for HR personnel, managers and employees, ensuring efficient management and transparency across all operations.

HR can manage employee data, handle leave approvals, track attendance and post announcements or news. Employees can apply for leaves, maintain their personal records, mark attendance and stay updated with announcements. Managers oversee leave requests and manage training and development programs for their teams. The platform improves productivity, communication and workflow across departments by digitizing core HR functions.

1.1 Purpose

To replace paper registers and scattered spreadsheets with one secure, role-based platform where HR work is recorded once and visible to the right people.

1.2 Scope

- Employee records, leave workflow, daily attendance, announcements and training management.
- Three roles: HR, Manager and Employee, with JWT-based authentication.
- Responsive web interface that works on desktop and mobile browsers.

2. Ideation Phase

2.1 Define the Problem Statements

Organizations that run HR on paper or spreadsheets face delays in leave approvals, errors in attendance records and poor communication. The problem statements below describe the main user groups.

	PS-1 (HR)	PS-2 (Manager)	PS-3 (Employee)
I am	an HR executive	a team manager	an employee
I am trying to	keep employee records, attendance and leave data in one place	approve my team's leaves and plan trainings quickly	apply for leave and see my attendance and announcements
But	data is spread over files, registers and chats	requests reach me late and I cannot see who is present	I do not know the status of my request or latest notices
Because	there is no central digital system	there is no team-wise view	updates are shared informally
Which makes me feel	overloaded and error-prone	uninformed	uncertain and ignored

2.2 Empathy Map (HR Executive)

Says	Thinks
"Who is on leave this week?" "I need the attendance report today."	Manual records are risky. Everyone should see the same data.
Does	Feels
Collects leave requests on paper and chat; updates spreadsheets by hand; posts notices on notice boards.	Stressed by repetitive work; worried about mistakes; wants transparency.
Pains	Gains
Lost requests, duplicate data, no history, delayed approvals.	One dashboard, instant approvals, clear history, quick announcements.

2.3 Brainstorming and Idea Prioritization

Ideas were collected for each user group and ranked by impact and effort. Ideas marked High priority were implemented in the first release.

Idea	Impact	Effort	Priority
Secure login with role-based access (HR / Manager / Employee)	High	Medium	High
Employee database with add / edit / delete	High	Low	High
Online leave application and approval with comments	High	Medium	High
Daily check-in / check-out attendance	High	Low	High
Announcements and news board	Medium	Low	High

Idea	Impact	Effort	Priority
Training programs assigned to team members	Medium	Medium	High
Role-wise dashboard with live counts	Medium	Low	High
Payroll and payslips	High	High	Future
Email / SMS notifications	Medium	Medium	Future
Reports export (PDF / Excel)	Medium	Medium	Future

3. Requirement Analysis

3.1 Customer Journey Map

Stage	Employee	Manager	HR
Login	Signs in with email and password	Signs in and lands on team dashboard	Signs in and sees company dashboard
Daily work	Checks in, later checks out	Views team attendance	Views attendance of everyone by date
Leave	Applies for leave, tracks status	Reviews team requests, approves or rejects	Reviews any request, filters by status
Growth	Sees assigned trainings	Creates trainings and assigns team	Creates and manages all trainings
Communication	Reads announcements	Reads announcements	Posts, edits and deletes announcements

3.2 Solution Requirements

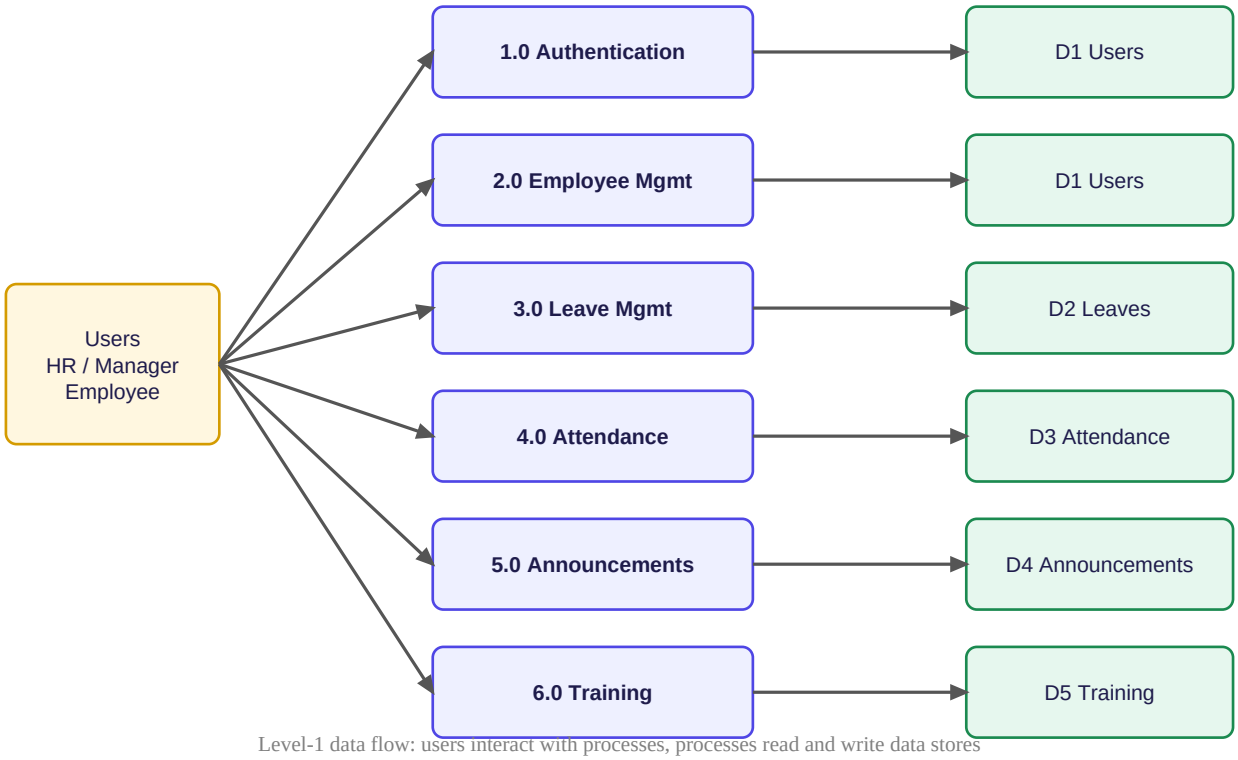
Functional requirements

- FR-1 User login with JWT, logout, and password change.
- FR-2 HR can create, update, deactivate and delete employees and assign a reporting manager.
- FR-3 Employees and managers can apply for leave; managers approve team leaves and HR approves any leave.
- FR-4 One attendance record per user per day with check-in, check-out and working hours.
- FR-5 HR can post, edit and delete announcements; everyone can read them.
- FR-6 Managers and HR can create trainings and assign them to employees.
- FR-7 Role-wise dashboard statistics.

Non-functional requirements

- Security: passwords hashed with bcrypt, JWT expiry, role-based authorization on every API.
- Usability: responsive Bootstrap interface with clear forms and messages.
- Reliability: validation on the server, unique attendance per day, duplicate-email protection.
- Maintainability: separate client and server folders, modular routes and models.

3.3 Data Flow Diagram



Level-1 data flow: users interact with processes, processes read and write data stores

3.4 Technology Stack

Layer	Technology
Frontend	React.js (Vite), React Router, Bootstrap 5, Axios
Backend	Node.js, Express.js (REST API)
Database	MongoDB with Mongoose
Authentication	JSON Web Tokens (JWT), bcrypt password hashing
Tools	VS Code, Postman, Git and GitHub

4. Project Design

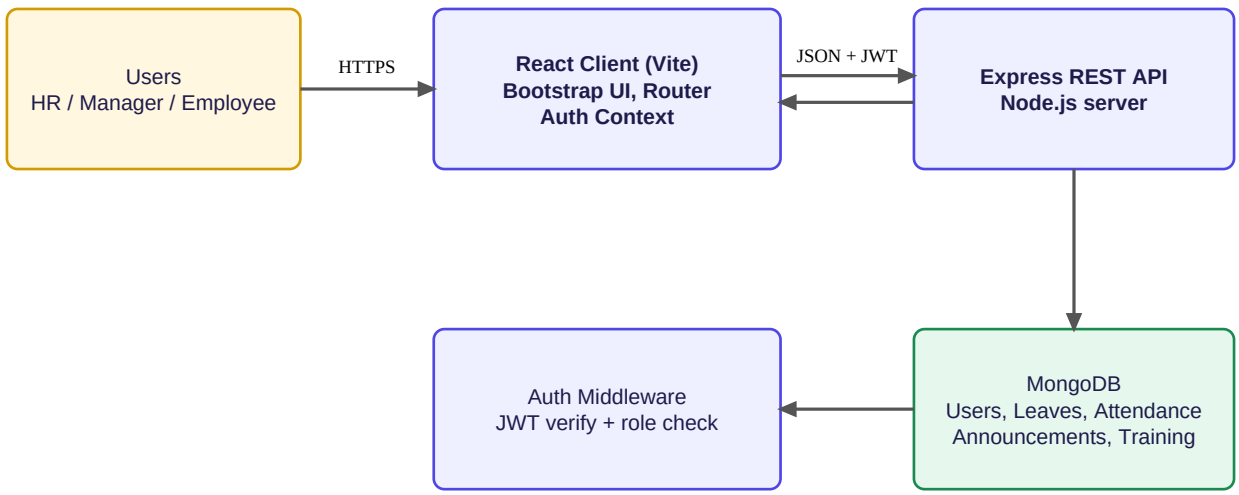
4.1 Problem-Solution Fit

Problem	Solution in StaffSync
Paper and spreadsheet records are scattered	Single MongoDB database with a web interface for every role
Leave approvals are slow and untracked	Online application, status tracking, approval with comments
Attendance errors and no visibility	One-click check-in / check-out with daily team and company view
Notices do not reach everyone	Central announcement board visible after login
Anyone can see everything	JWT login with HR / Manager / Employee permissions

4.2 Proposed Solution

A React single-page application consumes a secured Express REST API. HR owns master data and announcements, managers act on their own team's leaves and trainings, and employees manage their own leave, attendance and profile. Dashboards show live numbers for each role.

4.3 Solution Architecture



Every protected request passes through JWT and role-based checks

4.4 Database Design

Collection	Main fields
User	name, email, password (hashed), role, department, designation, phone, address, dateOfJoining, salary, manager, isActive
Leave	employee, leaveType, fromDate, toDate, days, reason, status, reviewedBy, reviewComment
Attendance	employee, date, checkIn, checkOut, workHours, status (unique per employee per day)

Collection	Main fields
Announcement	title, message, type (Announcement / News / Policy / Event), postedBy
Training	title, description, trainer, mode, startDate, endDate, assignedTo, createdBy

REST API Overview

Endpoint	Method	Access
/api/auth/login	POST	Public
/api/users	GET / POST / PUT / DELETE	HR (list also for Manager)
/api/leaves	POST, GET, PUT status	Employee applies; HR / Manager review
/api/attendance	POST checkin / checkout, GET	All users; HR / Manager view team
/api/announcements	GET / POST / PUT / DELETE	Read: all; write: HR
/api/training	GET / POST / PUT / DELETE	HR / Manager create; Employee views
/api/dashboard/stats	GET	All, role-based numbers

5. Project Planning

The work was divided into 6 epics and delivered in four sprints.

Sprint	Epic	User stories / tasks
Sprint 1	Setup and Authentication	Project setup (client and server); MongoDB connection; User model; login with JWT; role-based middleware
Sprint 2	Employee and Leave Management	Employee CRUD for HR; apply for leave; approve / reject with comments; cancel pending leave
Sprint 3	Attendance and Announcements	Check-in / check-out; attendance history and team view; announcements CRUD; news feed
Sprint 4	Training, Dashboard and Profile	Training programs and assignment; role-wise dashboard; profile and password change; testing and documentation

6. Setup and Execution

Requirements: Node.js 18+, npm, MongoDB, Git, Visual Studio Code.

```
# Backend
cd server
npm install
npm run seed      # demo data
npm run dev       # http://localhost:5000

# Frontend (new terminal)
cd client
npm install
npm run dev       # http://localhost:5173
```

Role	Email	Password
HR	hr@staffsync.com	hr12345
Manager	manager@staffsync.com	manager123
Employee	employee@staffsync.com	employee123

7. Functional Testing

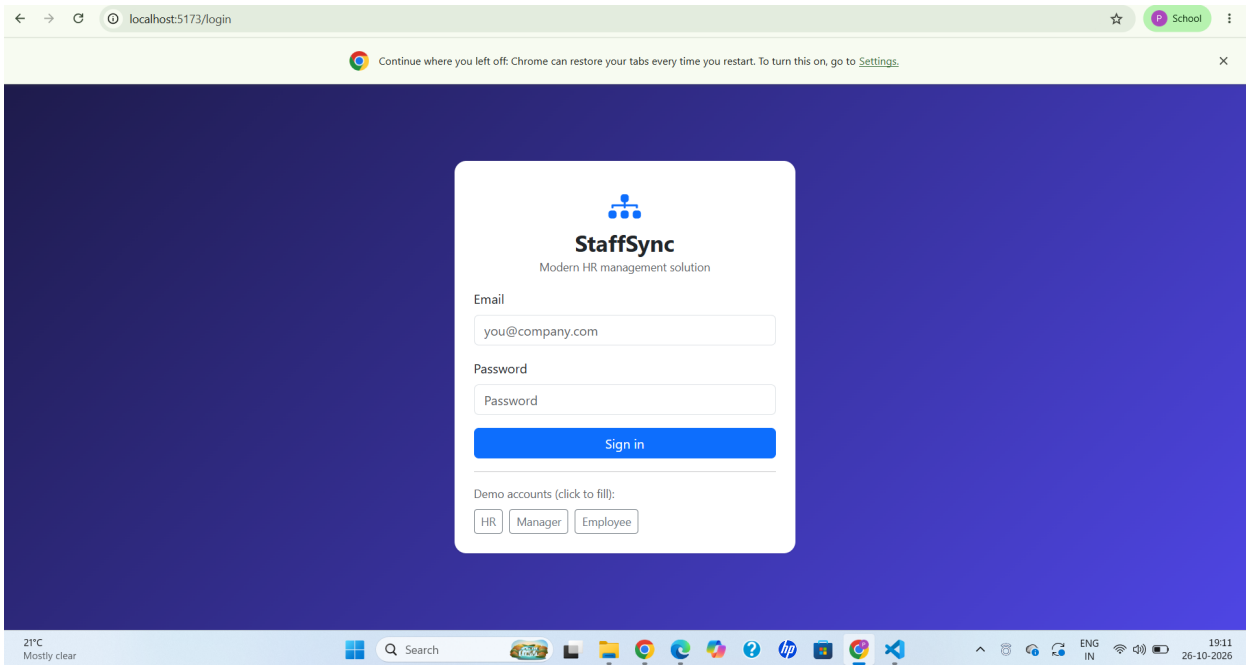
Manual functional testing was done on the running application using the demo accounts. Results correspond to the screenshots in the next section.

ID	Test case	Expected result	Result
TC-01	Login with valid HR / Manager / Employee credentials	User reaches role-specific dashboard	Pass
TC-02	HR opens Employees page	All employees listed with add / edit / delete	Pass
TC-03	HR opens Leaves page	All requests shown with status filters and Review action	Pass
TC-04	Manager opens Leaves page	Only own team's requests shown	Pass
TC-05	Employee opens My Leaves	Own leave requests with Cancel option	Pass
TC-06	Check In / Check Out	Time recorded; status shown in history	Pass

ID	Test case	Expected result	Result
TC-07	Open Announcements as HR / Manager	HR sees Post New; Manager sees read-only list	Pass
TC-08	Open Training as HR	Program with trainer, dates and assignees	Pass
TC-09	Open My Profile	Details shown; edit and change password form	Pass

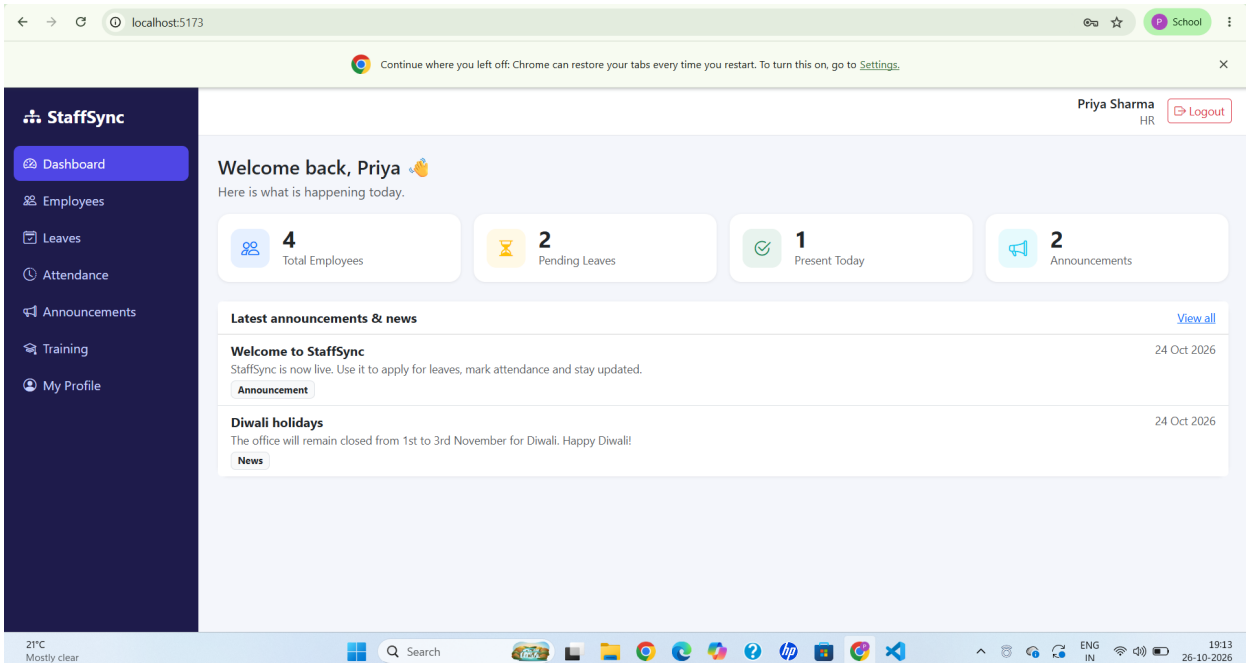
8. Results: Application Screenshots

Figure 1: Login Page



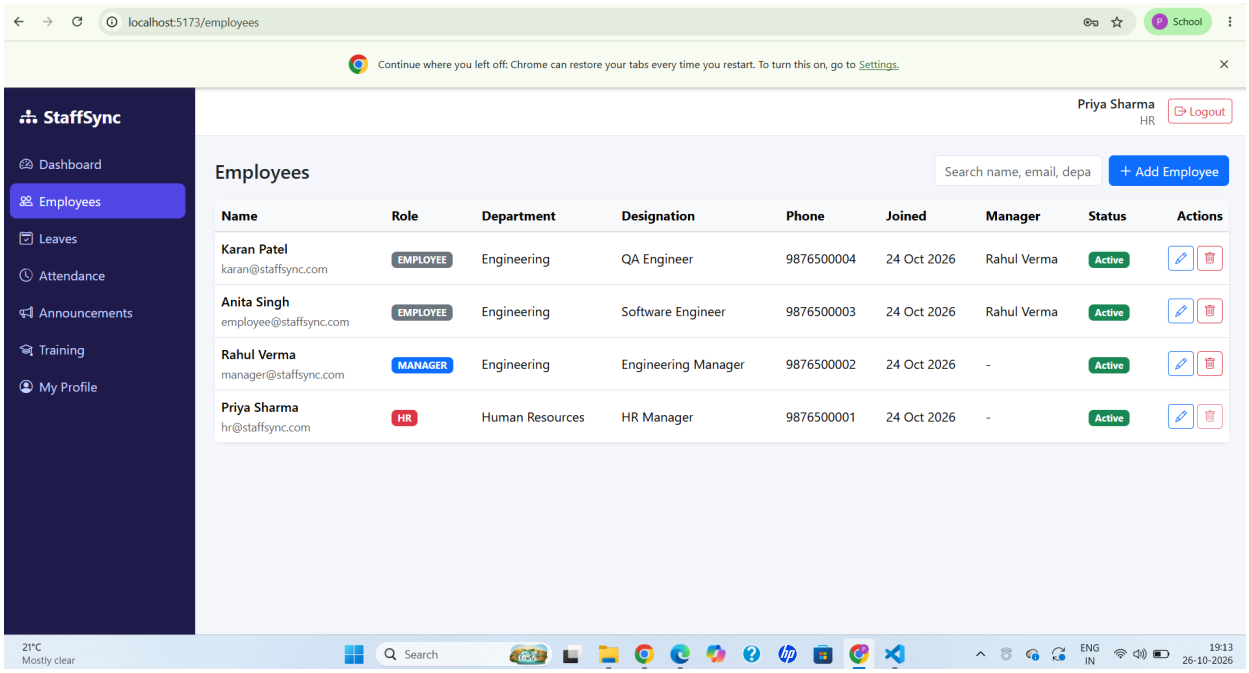
Common login screen for all roles. Email and password are verified and a JWT token is issued. Demo-account buttons fill the credentials quickly.

Figure 2: HR - Dashboard



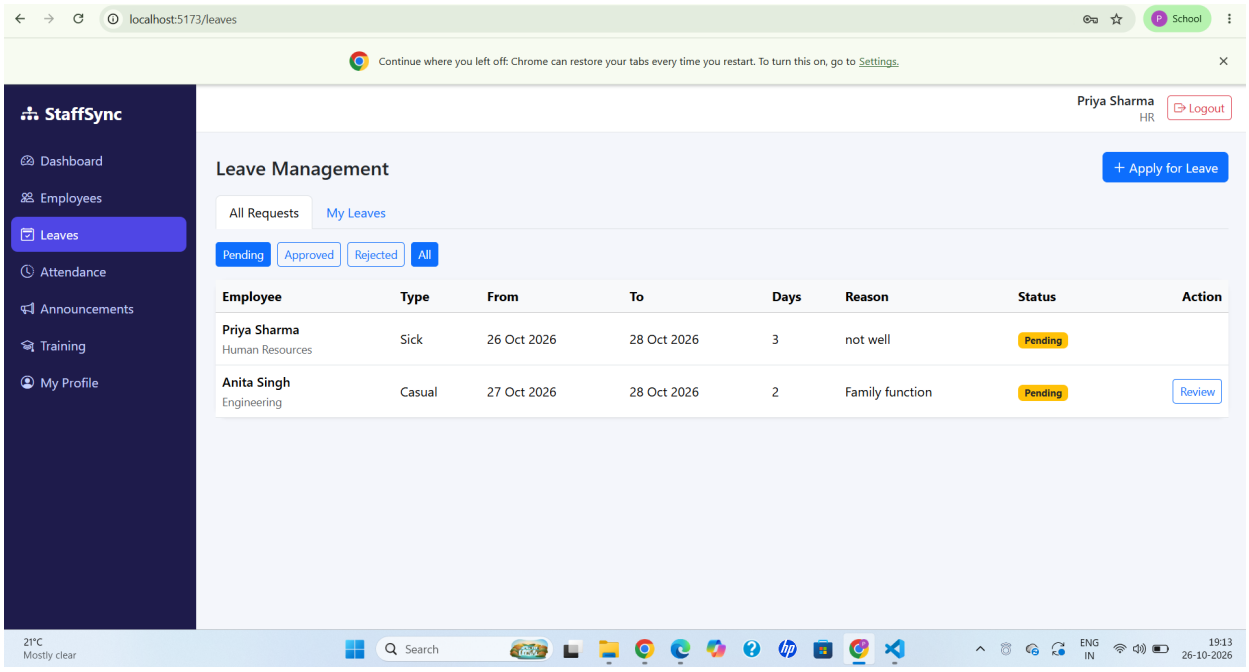
Summary cards for total employees, pending leaves, employees present today and announcements, plus the latest announcements.

Figure 3: HR - Employees



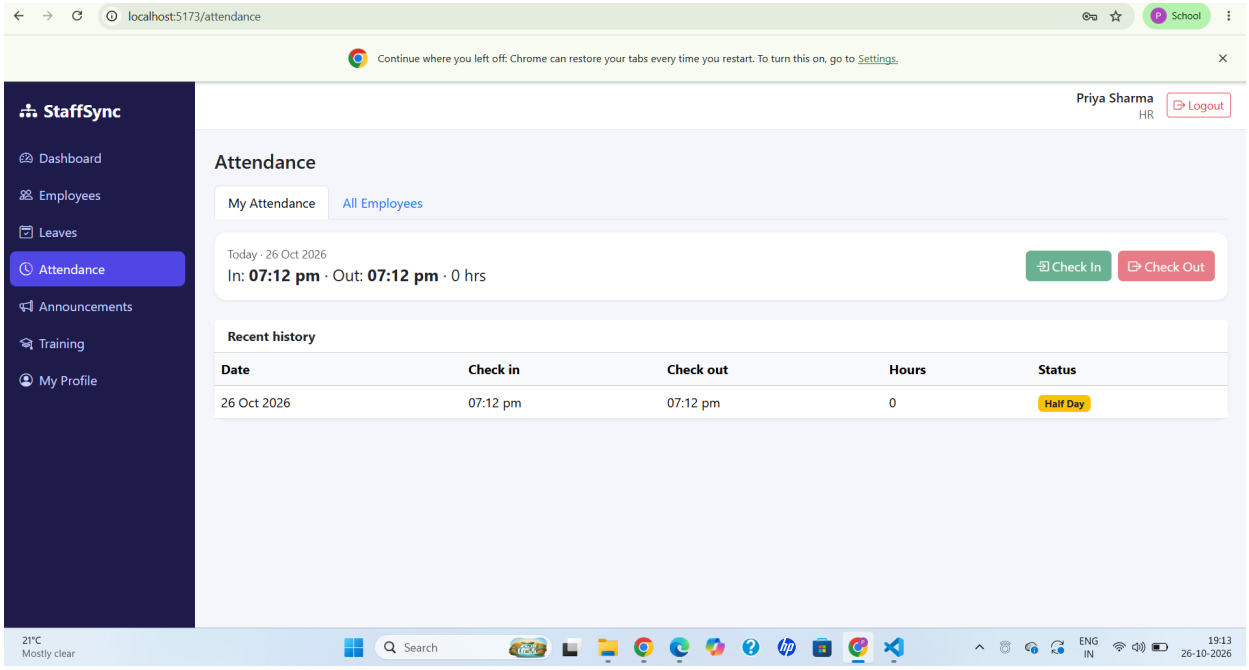
HR can view all employees, search them, add new employees, edit details and delete or deactivate accounts.

Figure 4: HR - Leave Management



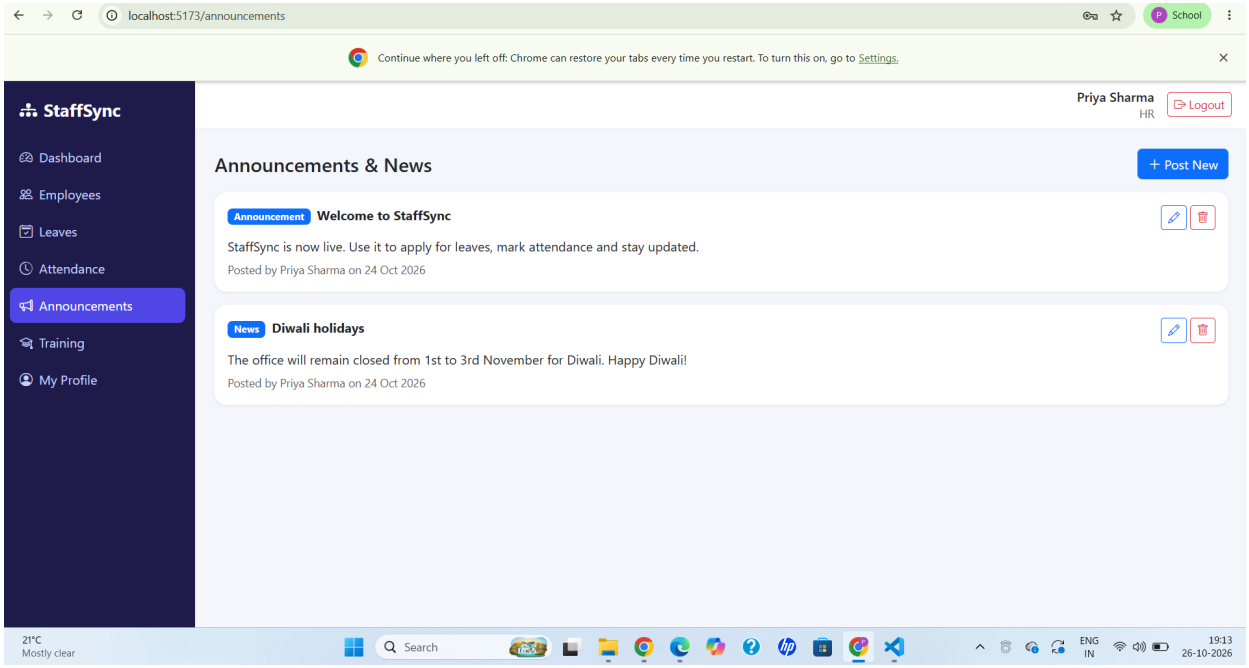
HR sees every leave request, filters by status (Pending, Approved, Rejected) and approves or rejects requests with a comment.

Figure 5: HR - Attendance



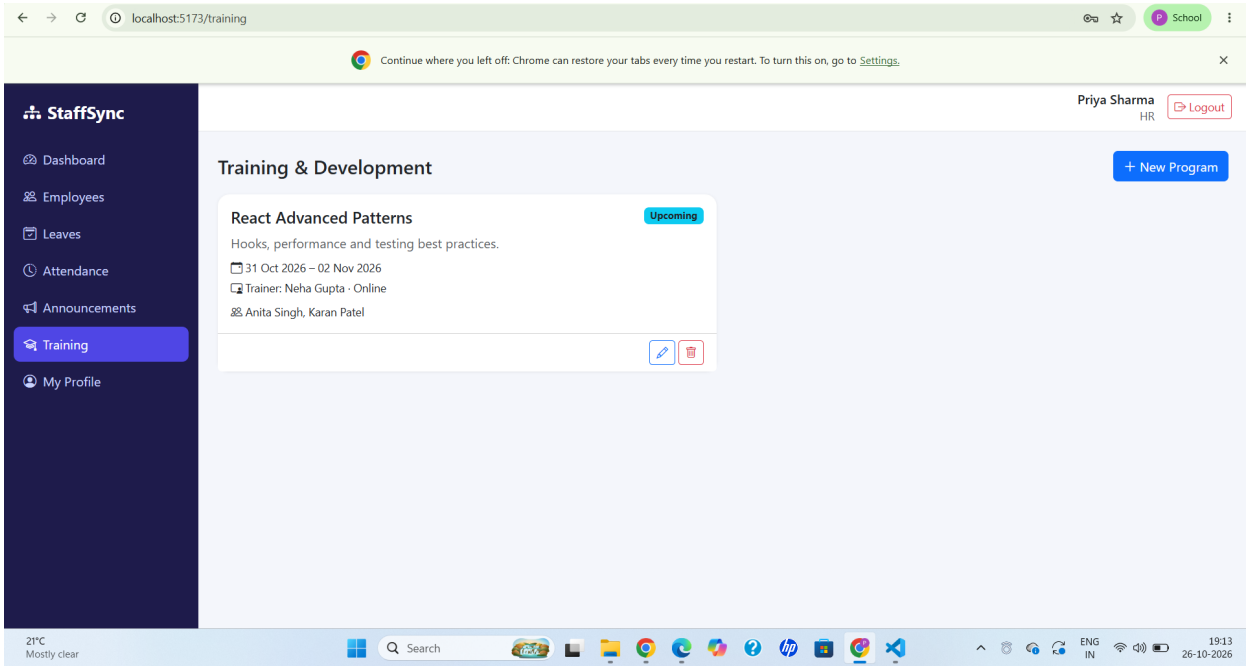
Check-in and check-out for the HR user with attendance history. The 'All Employees' tab shows company-wide attendance by date.

Figure 6: HR - Announcements



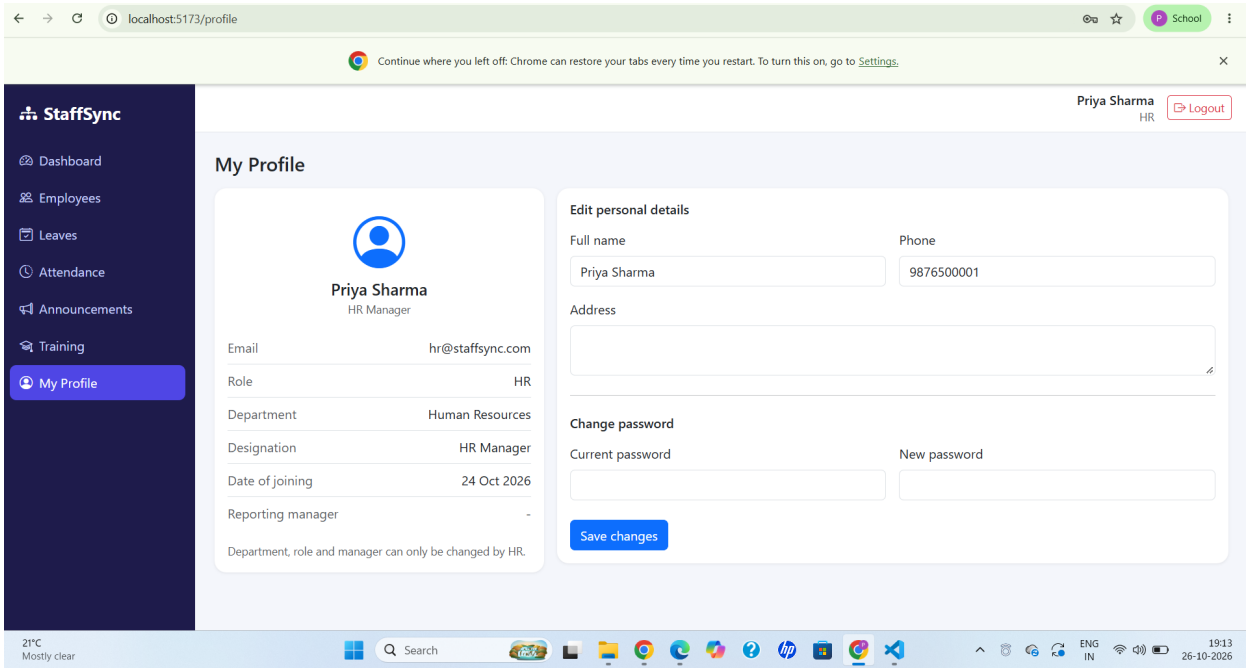
HR posts, edits and deletes announcements, news, policies and events that every user can read.

Figure 7: HR - Training



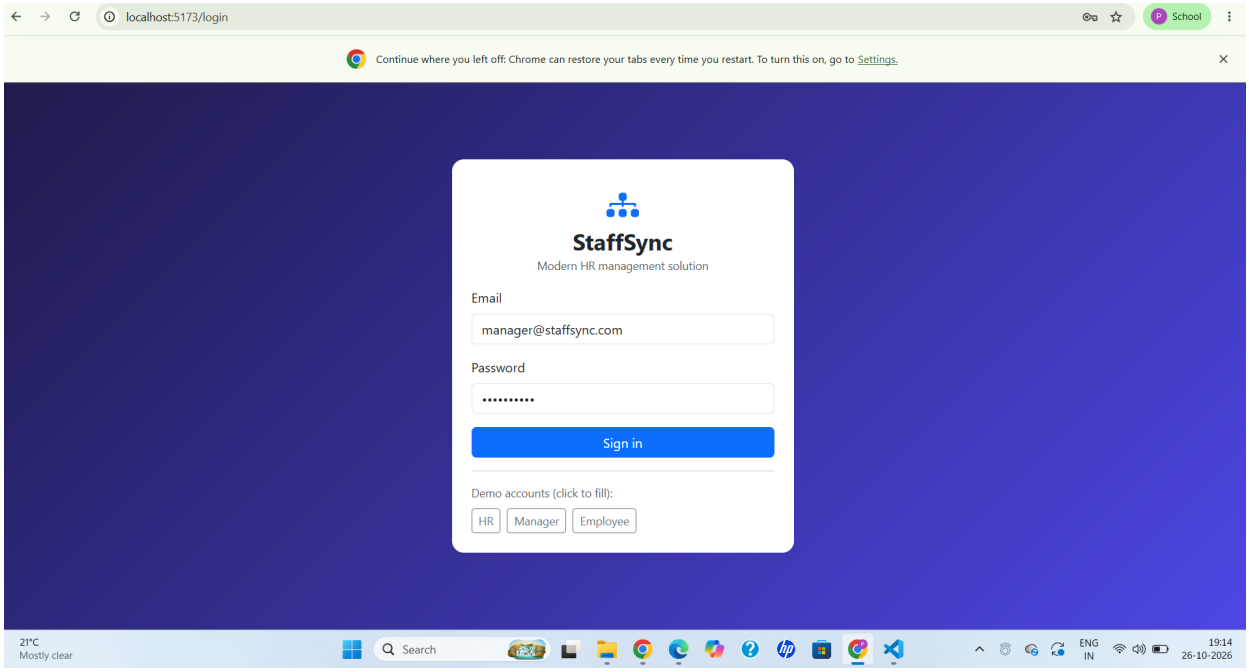
Training and development programs with trainer, mode, dates and the employees they are assigned to.

Figure 8: HR - My Profile



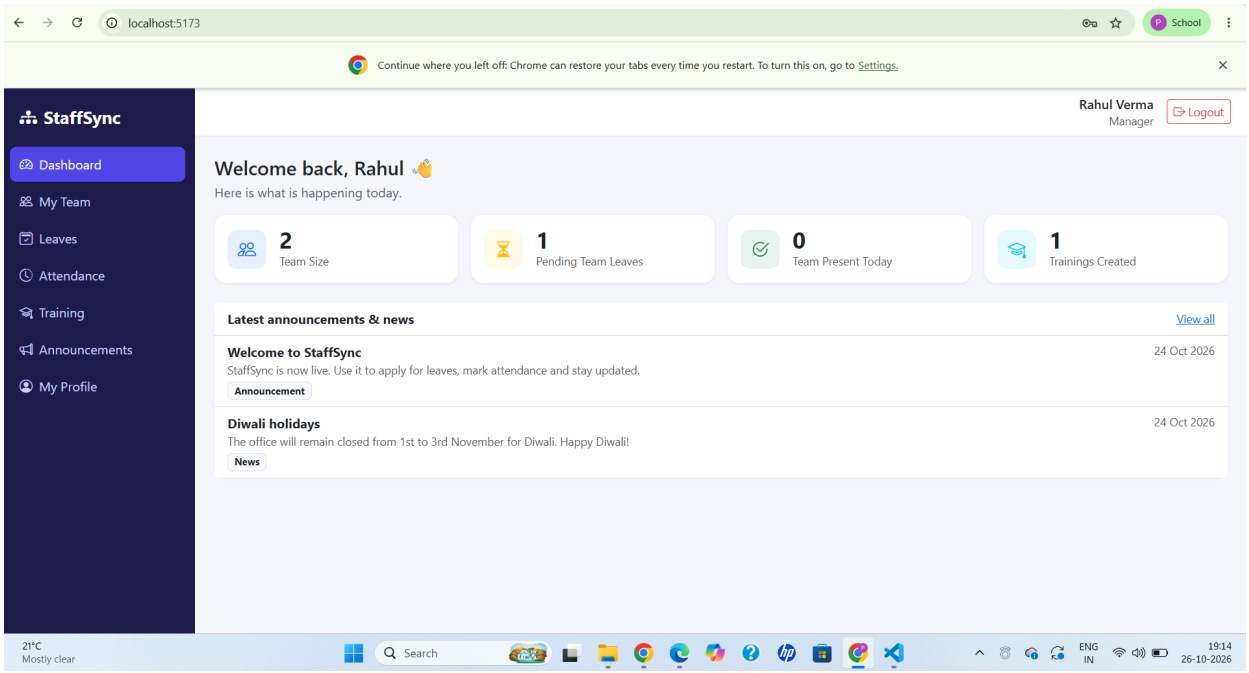
Users update their name, phone and address and change their password. Role, department and manager are controlled by HR.

Figure 9: Manager - Login



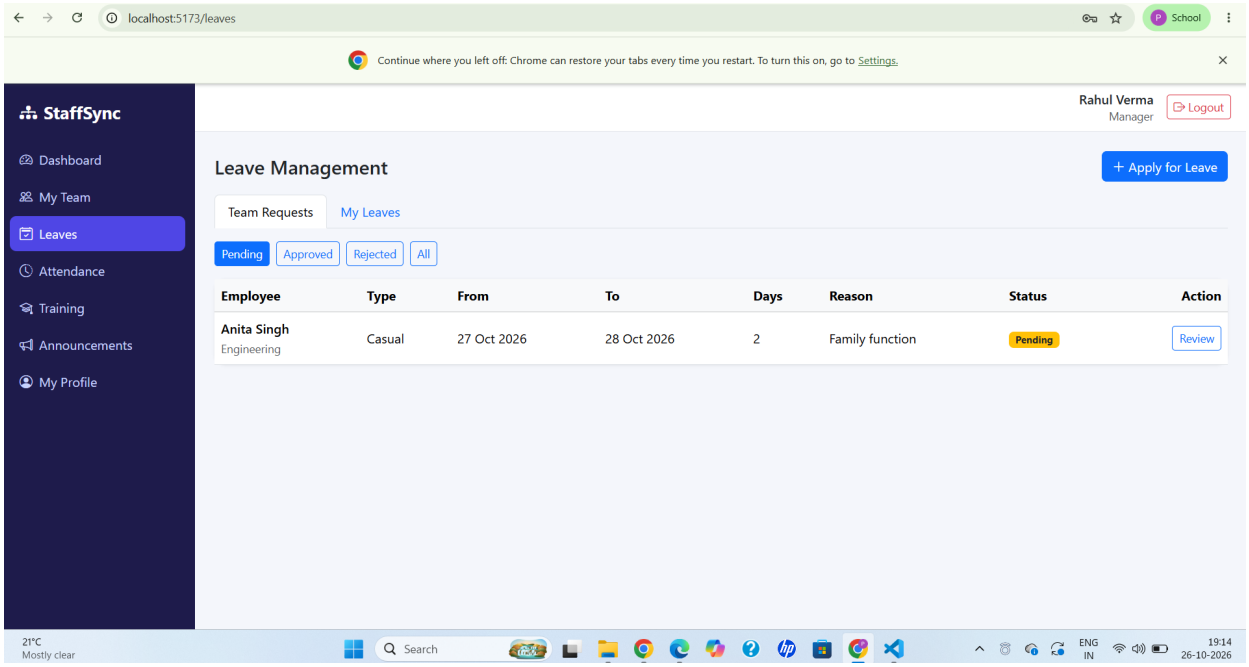
Login with a manager account.

Figure 10: Manager - Dashboard



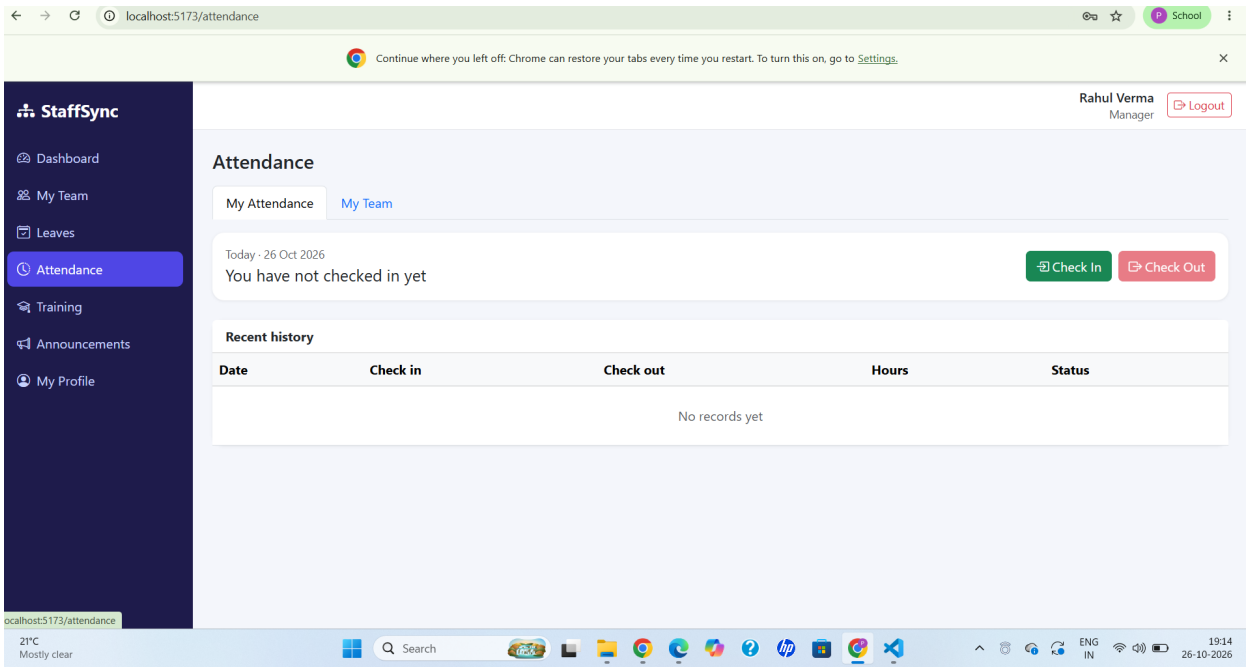
Team size, pending team leaves, team members present today and trainings created.

Figure 11: Manager - Team Leave Requests



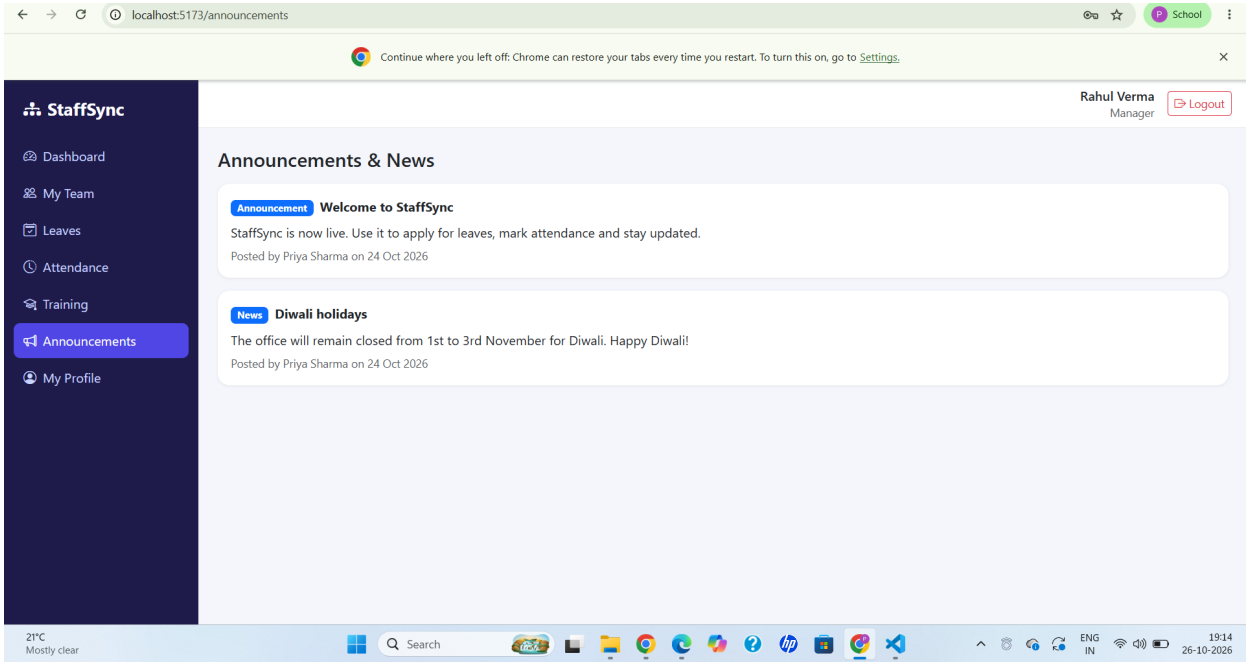
Managers review leave requests only from their own team and approve or reject them.

Figure 12: Manager - Attendance



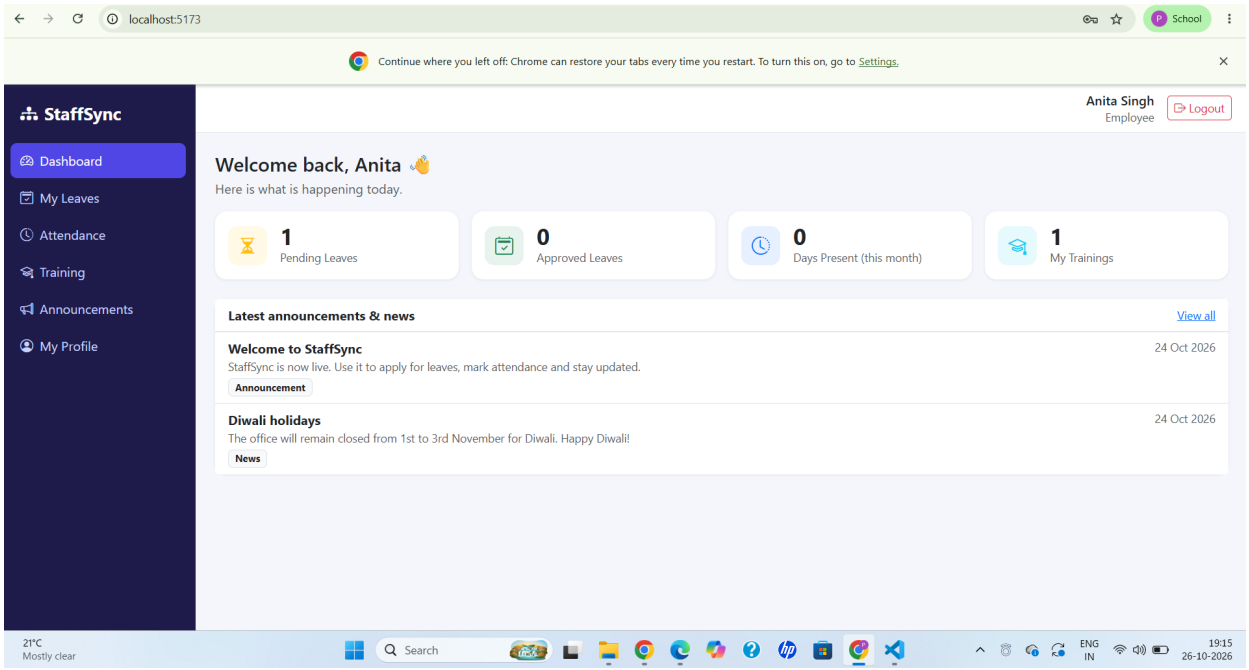
Managers mark their own attendance and view the attendance of their team.

Figure 13: Manager - Announcements



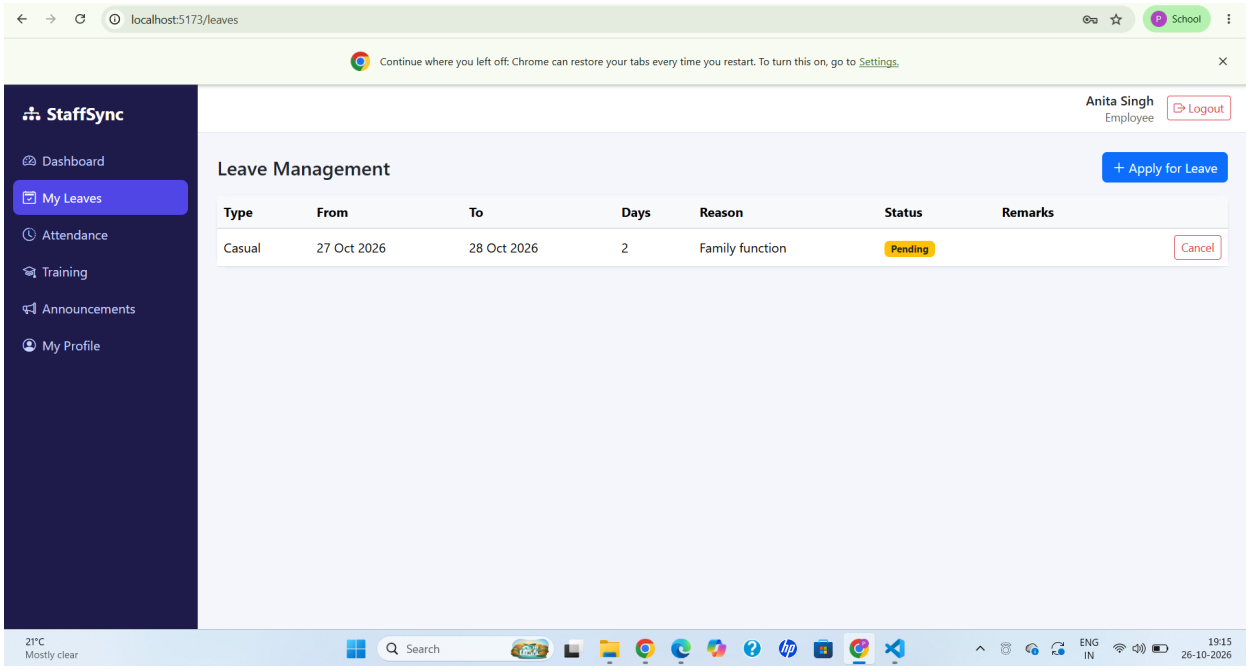
Managers can read company announcements and news (read-only).

Figure 14: Employee - Dashboard



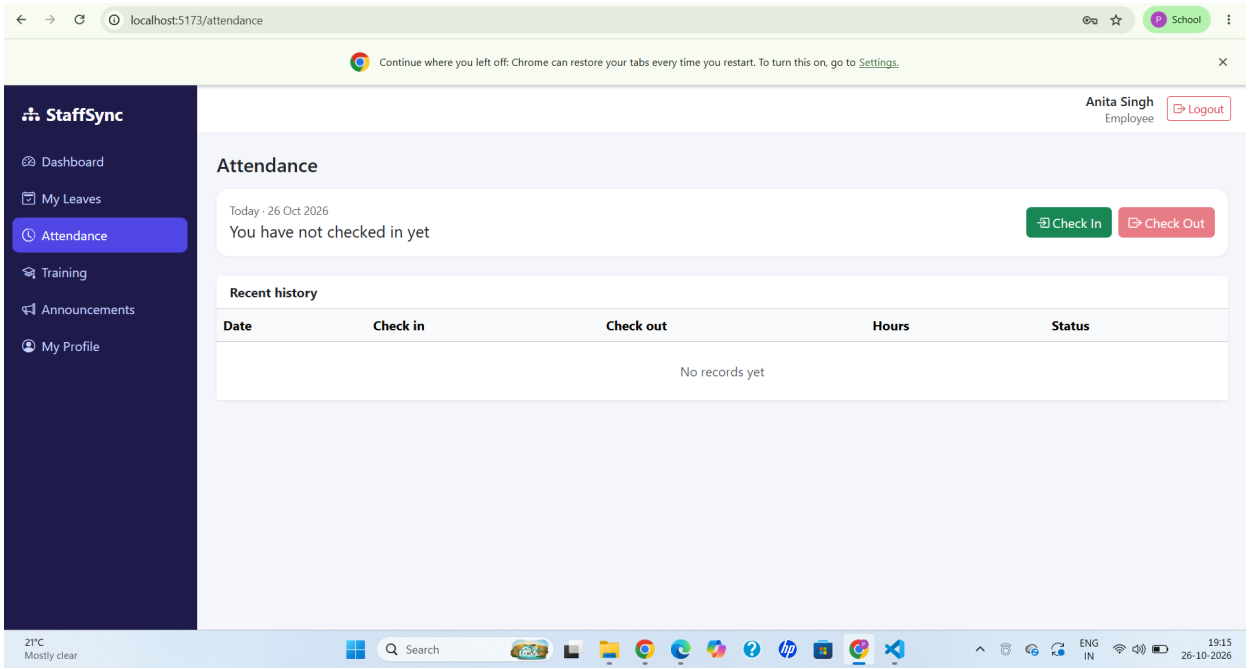
Pending leaves, approved leaves, days present this month and assigned trainings.

Figure 15: Employee - My Leaves



Employees apply for leave, track status and remarks, and cancel pending requests.

Figure 16: Employee - Attendance



Employees mark attendance with Check In and Check Out and see their recent history.

9. Advantages and Disadvantages

Advantages	Disadvantages
Centralized HR data for the whole organization Role-based security with JWT Faster leave approvals with full history Simple attendance with check-in / check-out Easy to extend because of modular code	Needs an internet / network connection and a running server No payroll, email notifications or reports export yet Attendance is self-marked (no biometric or location check) No leave balance limit per leave type yet

10. Conclusion

StaffSync centralizes HR operations into one platform. Role-based access keeps data secure, leave and attendance workflows are automated, and announcements and training keep everyone informed. The project demonstrates a complete MERN stack application with authentication, authorization, REST APIs and a responsive interface.

11. Future Scope

- Payroll and downloadable payslips.
- Email / SMS notifications for leave decisions and announcements.
- Leave balance and holiday calendar.
- Reports and analytics with PDF / Excel export.
- Biometric or geo-fenced attendance.
- Cloud deployment (Vercel / Render / MongoDB Atlas) and a mobile app.

12. Appendix

Item	Details
Source code (GitHub)	https://github.com/en24cs3010799-cmd/staffsync-hrms
Frontend	client/ (React + Vite + Bootstrap)
Backend	server/ (Node.js + Express + MongoDB)
Run locally	Frontend: http://localhost:5173 Backend: http://localhost:5000